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 Creation date: 25 April 2026
 Modification date: 14 May 2026
 Project(s) related: E-Learning System Ontology(v2)
 Implementation Language OWL 2 (Turtle Syntax)

Ontology Requirements Specification Document	
1	Purpose
	The purpose of this ontology is to semantically model the academic processes in an e-learning platform (enrollment, teaching, assessment, and performance tracking) and to reveal hidden relationships between data through logical inference. Version 2 is based on the separation of TBox (Schema) and ABox (Data) in order to ensure system modularity and data independence.
2	Scope
	The ontology covers the core academic lifecycle within a university or MOOC platform. Included: People (Student, Instructor), Academic Content (Course, Module), Assessment and Evaluation (Assignment, Submission, Grade), and logical prerequisite chains. Excluded: User authentication (passwords), financial transactions, video storage infrastructure, and social communication tools (chat/forums).
3	Implementation Language (optional)
	This ontology has been developed in compliance with the W3C standards using OWL 2 (Web Ontology Language). To improve readability and modularity of the code, Turtle (.ttl) syntax was preferred instead of XML/RDF. The Protégé editor was used for development and logical inference (reasoning) testing.
4	Intended End-Users (optional)
	* Academic Administrators: To monitor student performance and course workloads. * System Developers: To integrate the ontology as a backend for a chatbot or intelligent recommender system. * Instructors: To semantically relate course materials with student performance.
5	Intended Uses
	* Automatic Prerequisite Validation: Determining whether a student is eligible to enroll in a course through a TransitiveProperty prerequisite chain.

	* Performance Analysis: Identifying students with low grades and overdue assignments using a reasoning engine. * Data Population: Automatically generating ontology individuals (instances) from unstructured course syllabi with the help of an LLM (Large Language Model).
6	Ontology Requirements
	a. Non-Functional Requirements
	Modularity: Physical separation of TBox (elearning-tbox.ttl) and ABox (elearning-abox.ttl) files. Readability: Use of rdfs:label and rdfs:comment annotations for all entities. Standards Compliance: Full compliance with W3C OWL 2 standards and Turtle syntax.
	b. Functional Requirements: Lists or tables of requirements written as Competency Questions and sentences
	These questions represent logical requirements that test the success of the ontology and demonstrate its difference from a simple database: CQ1: Who is the instructor of a specific course, and which other courses does this instructor teach? (Tested logical feature: InverseOf and FunctionalProperty) CQ2: What are all the direct and indirect (transitive) prerequisites of a course? (Tested logical feature: TransitiveProperty chains) CQ3: Which students are not enrolled in at least one course, violating system rules? (Tested logical feature: class restrictions – minCardinality [1]) CQ4: What is the complete list of students in all courses taught by a specific instructor? (Tested logical feature: Property Chain – isTaughtBy → hasStudent) CQ5: Which students submitted an assignment after the deadline (late submission)? (Tested logical feature: Datatype Reasoning / date-time comparison)
7	Pre-Glossary of Terms (optional)
	a. Terms from Competency Questions
	Person: Student, Instructor AcademicContent: Course, Module Assessment: Assignment Evaluation: Grade, Score Submission: Student's submitted work
	b. Terms from Answers
	enrolledIn / hasStudent: Inverse relationship between student and course. hasPrerequisite: Transitive prerequisite relationship between courses. teaches / isTaughtBy: Teaching responsibility relationship between instructor and course. submits / submittedBy: Relationship between student and submitted work.

	c. Objects
	Individuals: Course_KnowledgeEng, Instructor_ProfessorSmith, Student_JalilGuliyev, Grade_A.